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ABSTRACT

The rapid population growth in Egypt necessitates innovative, sustainable approaches to increase the national agricultural base. The purpose of the study is to expand agricultural regions into desert-like conditions. As the population increased to 119 million, the corresponding per capita consumption rate created enormous demand that bears no resemblance to local crop production. Evidently, Egypt cultivates approximately $4,058 \times 10^3$ hectares; Egypt secures 37th place in Africa. Narrowing the scope to the 6th October city, it is a city with a population that is estimated at more than 670 thousand people. Despite this high population, it is coupled with desert-like conditions: soil moisture is 1%, and high temperatures. Furthermore, they consume more than 0.5 million acres. Thus, the problem is addressed via an even-span smart greenhouse. This project aims to ensure an optimal, sustainable environment for crops such as orchids and coriander. The small-scale prototype is constructed in an area that doesn't exceed 0.25 m², with a feedback system hosted by ESP-32. The system gathers readings for the soil moisture, light intensity, and temperature; meanwhile, they are synchronized. It then compares these real-time metrics against conditions needed for selected crops individually, automatically compensating for any disturbance. This prototype can work for 6 continuous hours over 5 consecutive days, optimizing the environmental conditions. It can effectively operate, as indicated by measured results: 22.85 °C temperature, 1686 soil moisture, 57.08% humidity, and 785.24LDR.

INTRODUCTION

The significant agricultural deficit in Egypt is driven by a rapid population growth crisis. This imbalance between the domestic production and food demand is especially prominent in Egyptian crop imports, which reach approximately 23-25 million tons annually (United States Department of Agriculture, 2025). Another major contributor is the scarcity of arable land in Egypt (nearly 3.1%), which stands below the global average of around 14.3% (Figure 1) (World Bank Open Data, 2022). These issues are coupled with the limited recycling infrastructure; Egypt secures the 7th place on the plastic mismanagement list (Simpson, 2024). This further exacerbates soil pollution. These constraints particularly appear in the city of 6th October, a desert-like region. Evidently, the average temperature is 36 °C, with low precipitation of 0.1 mm. Despite the harshness, it has more than 600,000 residents. This enormous population consumes about 150-200 ft³ per capita (Lisa2025). At the same time, the city of 6th October produces a tiny percentage of its crop consumption.

Similar issues were addressed via solutions such as Azahir Farm, which relies heavily on recycled, old components in its construction. This project is effective as it utilizes an adaptive mechanism that effectively prevents sudden environmental fluctuations; however, it poses some power issues. Furthermore, the Jain Irrigation system, which implements micro-irrigation. This approach drives water directly into plant roots through small tubes and emitters, minimizing waste of water. Nevertheless, it has limited repair flexibility. In order to obtain the best results, the even-span smart greenhouse monitors three parameters: soil moisture, light intensity, and temperature. The greenhouse adopts an even-span structure, consisting of two roofs with equal slope and length. The structure ensures efficient sunlight distribution and proper air circulation, which regulates the internal temperature. Moreover, it provides fresh carbon dioxide for plants. The small-scale prototype should be constructed in an area that doesn't exceed 0.25 m², corresponding to dimensions of 0.4 m × 0.6 m. The entire system is an ESP-32-based closed-loop feedback control system. The system synchronizes the sensors' readings. It then compares the real-time metric against conditions needed for coriander and orchid cultivation. It automatically compensates for any disturbance during the 6-hour continuous operation, fulfilling the design requirements.

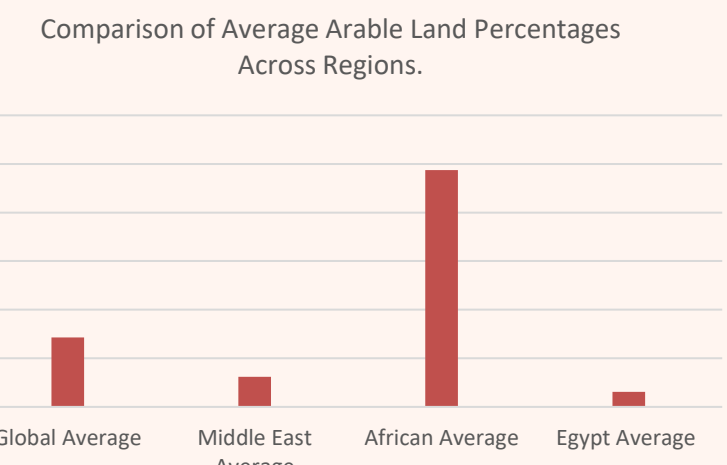


Figure 1: demonstrates the average arable land

MATERIALS & METHODS

Table 1: Materials Utilized in Prototype.

Materials	ESP 32	LDR Sensor	DHT-22 Sensor	Soil moisture sensor	Recycled-Bottles	Relays
Description	The controller board that used for controlling feedback system	used to detect the light and gives feedback for ESP 32	used to measure temperature and Humidity	Used to measure the humidity of Soil	Used a transparent cover to allow sunlight passing.	Used to control the operation of Actuator
Quantity	1	1	1	1	30	1
Image						
Materials	Beech Wood	Fans	Nails	White Pumps 5V	Power Supply	Jumper wires
Description	to construct the frame of our Prototype	to cool weather condition around plant	to connect the frame of prototype	to pump water inside prototype	to emerge power inside the circuit	used to connect components of circuit
Quantity	8 meters	3	50	2	1	60
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Methods:

1. Prototype design and sketching:

The project started with visualization and selecting the dimensions to make a 3D model using SolidWorks, as shown in Figure 2.

2. Electronic System Integration:

The control system was built using an ESP32 microcontroller mounted on a breadboard, including relay modules to safely handle high-voltage components. This is alongside a 12V power supply to operate actuators. All components were interconnected with ESP32, as shown in Figure 3, ensuring a robust electronic architecture for the automated operations.

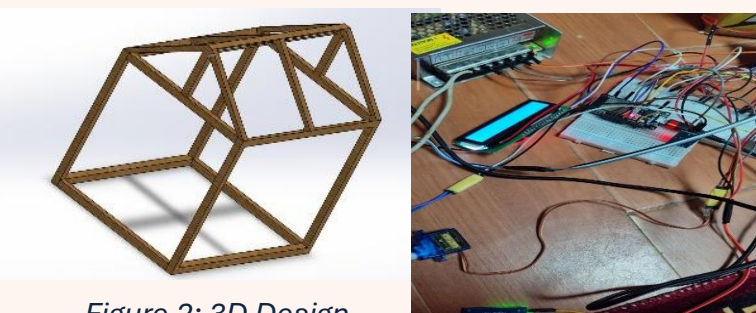


Figure 2: 3D Design

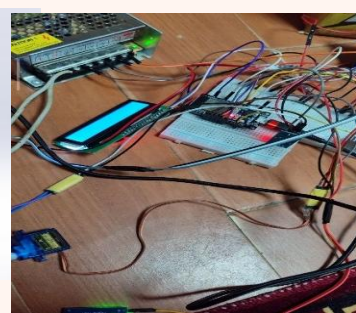


Figure 3: Electronic System

3. Sensors and actuators integration:

A DHT22 sensor was connected to monitor temperature; the ESP was programmed to activate a cooling fan when the temperature threshold was exceeded. Regarding the soil moisture sensor, it was utilized to activate a secondary water pump until the soil took the required water, as shown in Figure 4. Also, a Light Dependent Resistor (LDR) was used to monitor light intensity to control a 220V to provide artificial lighting, while a DC motor driven by an H-bridge module was implemented to operate a shading system.



Figure 4: Sensors & Actuators

4. Programming system:

An ESP32 code was programmed using Arduino IDE to control sensors and actuators, as shown in Figure 5. The system continuously reads data and activates corresponding actuators to maintain optimal environmental conditions automatically.

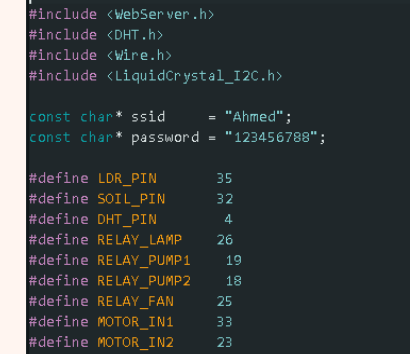


Figure 5: Programming

5. System testing and calibration:

To ensure the accuracy of the data collected from sensors, the calibration was conducted. The temperature data from the DHT22 sensor was validated by comparing the reading against an external digital thermometer. For the soil moisture sensor, readings were recorded in both completely dry and fully saturated soil to handle the water pump. This physical calibration ensured that the logic responded accurately to actual environmental conditions.

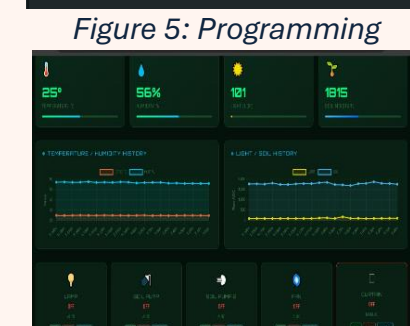


Figure 6: Website control

6. Web-Based Monitoring and Control:

A web-based interface was integrated and connected to the ESP32 via Wi-Fi. This website allows real-time monitoring of parameters and data, and allows manual control of the system components, which provides flexibility with automated operation and emergency cases, as shown in Figure 6.



Figure 7: Final Assembly

7. Assembly of the whole proto

A wooden frame was constructed using a combination of fabricated and recycled materials to form a box structure as a container for soil, as shown in Figure 7. Also, recycled plastic bottles were flattened to serve as a cover.

Test Plan:

A comprehensive set of criteria for testing was established to ensure the feasibility of the prototype to meet design requirements:

- The prototype was tested to ensure continuous operation for at least 6 hours without human intervention. The structure was verified to be assembled from recycled components. Moreover, sensors were calibrated using standard methods to ensure that accuracy does not exceed the ±5%.
- The ESP32 controller was tested to ensure the correct implementation of programming logic like IF-THEN rules, and the system was tested under different conditions to evaluate its ability to detect changes. Also, the system and circuit were tested to ensure stable power, voltage, and reliable performance.

RESULTS

Negative results

Initially, the system had noticeable delay and unstable behavior in the circuit, such as vibration-like fluctuations. After testing, the issue was caused by the water pump because it was connected to 12 volts instead of 5 volts. The issue was solved by adding the voltage regulator to reduce the voltage to 5 volts.

The project was implemented using Arduino Uno as the controller. After connecting the circuit components, it became clear that the system required data monitoring on a website to make it easier for the user. So, the issue was solved by using an ESP32 instead of an Arduino Uno to allow integration with Wi-Fi.

Positive results

The sensor data indicate stable and controlled environmental conditions, with temperature, soil moisture, humidity, and light maintained within desired ranges. This confirms the effectiveness of the feedback control system as shown in blew Figures.

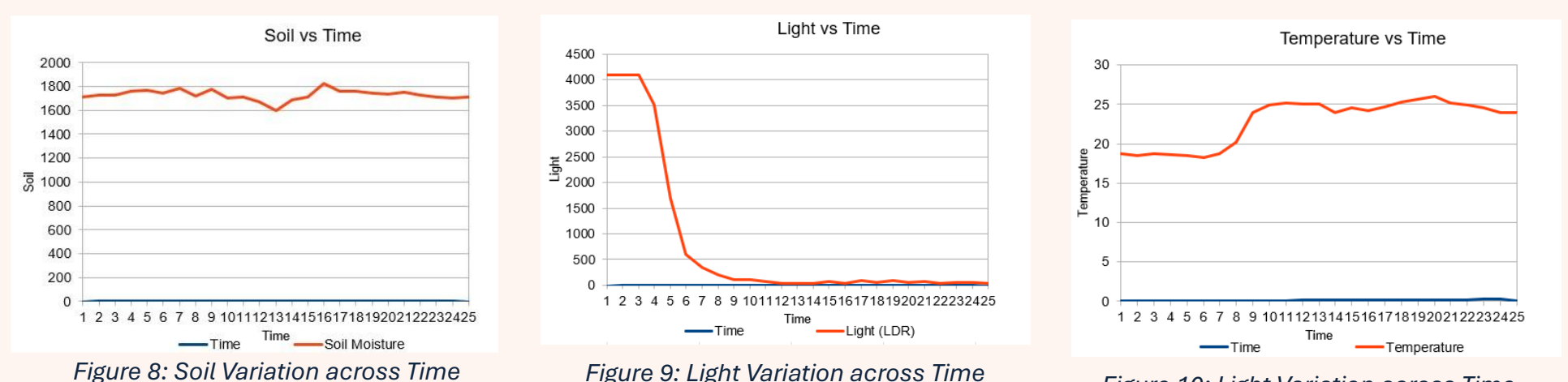


Figure 8: Soil Variation across Time

Figure 9: Light Variation across Time

Figure 10: Temperature across Time

Table 2: Parameters average value across each day for 5 consecutive days

Day Number	Temp	Humidity	soil	Light
Day 1	18.7	69.1	1713	4095
	25	52.5	1671	36
	22.2	65.7	1723	205
	18.2	68.5	1713	4095
Day 2	24.9	50.1	1671	36
	22	60	1723	205
	17.9	68.8	1718	4095
Day 3	24	52.2	1617	110
	20.5	40	1733	205
	18.9	69.1	1713	4095
Day 4	25.2	53.2	1671	32
	22.2	65.7	1723	205
	19	67	1728	4095
Day 5	25	52.5	1671	36
	23	65.7	1723	205

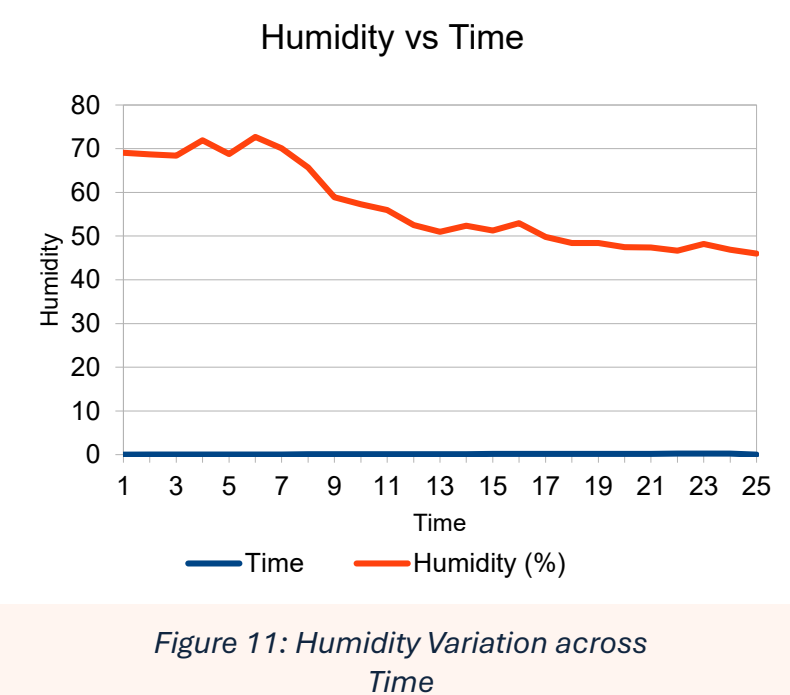


Figure 11: Humidity Variation across Time

ANALYSIS

After constructing the small-scale prototype, a test plan was conducted to verify that the prototype satisfies the design requirements. These criteria included cultivation area, which was measured to ensure that it does not exceed 0.25 m², corresponding to dimensions of 0.4 m × 0.6 m. Moreover, the system's endurance was evaluated through continuous operation for 6 hours per day over 5 consecutive days to emphasize its reliability. This performance resulted from an integration of electric and mechanical concepts.

Structural Engineering

The primary purpose of this investigation is to validate the system's capacity to sustain an optimal agricultural microclimate amidst the harsh, desert-like conditions of the 6th of October City. To achieve this purpose, the project's performance is evaluated across several interdisciplinary aspects:

At the outset, the extreme environmental loads of the region—elevated temperature (approximately 36.0 °C)—dictate strict architectural constraints. That is why the structure must satisfy two critical criteria: passive air circulation and climate optimization. In response to these constraints, the even-span greenhouse configuration emerges as the optimum solution, as illustrated in Figure (12). It is a greenhouse structure composed of two equal roof slopes that accommodates two to three rows of plant benches.

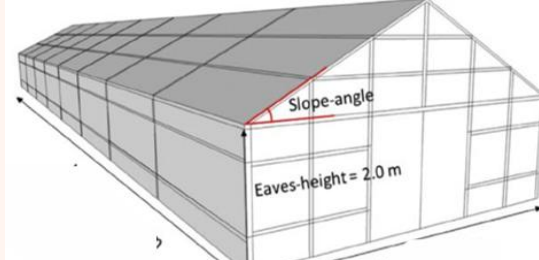


Figure 12: Even-Span Greenhouse Structure (Kwon et al., 2015)

Under desert-like conditions, the even span relies on several approaches that enable it to gradually decrease the temperatures:

First of all, the number of spans enhances the microclimate by regulating the air temperatures via smooth air paths. The 30° angle of inclination based on Bernoulli's principle (Equation 1):

$$\text{Equation 1: Bernoulli's Principle}$$

$$P_1 + \rho gh_1 + \frac{1}{2} \rho v_1^2 = P_2 + \rho gh_2 + \frac{1}{2} \rho v_2^2$$

To validate the aerodynamic efficiency of the structure, based on the average air velocity of the city of 6th October, which is estimated at 6.0 m/s (Simulated Historical Climate & Weather Data for 6th of October City - Meteoblue, n.d.). Determining the structural geometry at a 30° roof and applying Bernoulli's principle resulted in a negative static pressure zone of 11.5 pascals. This pressure differential acts as a natural extraction system; air naturally flows from regions of higher pressure to regions of lower pressure. In light of these considerations, the quantified Air Changes per Hour (ACH), which was modeled utilizing (Equation 2):

$$\text{Equation 2: Air Changes per Hour (ACH)}$$

$$ACH = \frac{Q}{V}$$

Where Q is defined as hourly airflow capacity per structural volume (V). This equation determines ventilation efficiency; the calculated airflow corresponds to 1.388 Air Changes per Hour.

Furthermore, the small-scale prototype has been constructed with a scale of 1:4.6 to the standard dimensions (Ponce et al., 2015); a width of 0.6 m, and a length of 0.4 m. Even more, the symmetric structure demonstrated in Figure 12 assists in distributing the load across the structural frame, which resulted in enhancing the structure's resistance to wind (ideal yearlong utilization) (Admin, 2025). This behavior arises from the symmetrical roof, which redistributes the aerodynamic loads via the configuration of pressure along the entire foundation. Thereby, it prevents localizing the stress, enhancing the overall structural stability.

Soil-Water Dynamics:

This study addresses the limited soil quality for sustainable cultivation in a hyper-arid soil profile. The city of 6th October is characterized by extremely low soil moisture (approximately 1%); meanwhile, the coriander and orchid require high, consistent moisture for survival. To verify this criterion, some quantitative analysis is conducted.

Primarily, the particle density is considered, with an average value of 2.66 g/cm³. This corresponds to a bulk density estimated at 1.4 – 1.8 g/cm³ (University of Nebraska, 2026). These parameters will be utilized to calculate the total soil porosity (Φ), according to the following formula that was established by Shreeja (2016) (Equation 3):

$$\text{Equation 3: Soil Porosity (Shreeja, 2016)}$$

$$\Phi = \left(1 - \frac{\rho_b}{\rho_p}\right) \times 100$$

Where ρ_b is bulk density, and ρ_p is particle density. This equation determines the porosity of soil; it corresponds to a range between 32.3 and 52.6%, depending on soil compaction. This indicates a relatively high concentration of macro-pores, which can rapidly percolate water.

Within this project, the coriander typically occupies a shallow root zone of approximately 0.01-0.02 m, particularly in sandy soils, allowing roots to absorb water (Modhavadia et al., 2025). Thus, the amount of water that would be pumped via the prototype's environmental optimization system is calculated utilizing the following mathematical model (Equation 4):

$$\text{Equation 4: Volume of water required for soil}$$

$$V_w = A \times Z \times (\theta_{FC} - \theta_i)$$

Where V_w is the total required water, Z indicates the effective root zone depth, and A is the prototype area. Furthermore, θ_{FC} and θ_i are the volumetric water content and the initial water content, respectively. The irrigation demand corresponds to approximately 0.5 L of water per irrigation cycle.

Feed-Back System

The observed stability in the prototype results is explained by principles of thermodynamic equilibrium and hydrological balancing in the system. The smart greenhouse uses a feedback system dependent on the ESP32 microcontroller to monitor environmental variables within defined conditions against external action perturbations, which is described in a schematic diagram as shown in Figure 13.

Thermodynamically, the system prevents thermal runaway by controlling the heat rate inside the greenhouse. When the DHT22 sensor detects the temperature exceeding the set threshold, the system alters the heat accumulation as shown in Equation 5:

$$\text{Equation 5: Net Thermal Accumulation Model}$$

$$q_{net} = q_{solar} - q_{conv}$$

Where q_{net} is the net heat transfer rate inside the greenhouse required to represent the thermal gain or loss of the system. q_{solar} is the incoming solar heat load, which represents the energy gain from solar radiation, q_{conv} is the heat removed from the system due to the cooling fan? The actuation of the DC motor to move the shade reduces the value of q_{solar}. Additionally, the cooling fan maximizes q_{conv}, which is represented by Newton's Law of Cooling (Equation 6):

$$\text{Equation 6: Newton's Law of Cooling}$$

$$q_{conv} = h \cdot A \cdot (T_{internal} - T_{ambient})$$

Where h is the convective heat transfer coefficient established by the fan's airflow, A is the effective internal surface area, and T represents the respective internal and ambient temperature. Hydrologically, the stability of soil moisture is analyzed through the conservation of mass principle. The soil moisture sensor triggers the water pump to compensate for evaporative losses, described by the following moisture balance equation 7:

$$\text{Equation 7: Differential mass balance equation}$$

$$\frac{d(\theta V)}{dt} = Q_{pump} - E_{rate}$$

Where θ is the volumetric water content of the soil (m³/m³), V is the control volume, Q_{pump} is the volumetric flow rate provided by the pump, and E_{rate} represents the evaporative losses rate from the system (m³/s). Furthermore, low-light conditions are corrected through an LDR sensor that activates a lamp, ensuring sufficient flux to maintain the baseline illuminance level for plant growth

The implementation of this active feedback control system is justified as it provides continuous real-time error correction compared to passive cooling or static timer-based irrigation systems, which cannot adapt effectively to dynamic environmental variables. By regulating q_{net} toward zero or negative values during peak thermal loads and adjusting the rate of change of $\frac{d(\theta V)}{dt}$ the system reduces environmental fluctuations. This closed-loop regulating mechanism helps explain the empirical results obtained, where the prototype maintained internal temperature within the target range of 15 °C to 25 °C and kept soil moisture within the optimal range of 40% to 50% under test conditions.

CONCLUSIONS

In light of these considerations, one major contributor to environmental pollution is the limited implementation of recycling, exacerbated by increasing population density, hindering Egypt's progress. These significant issues particularly appear in the city of 6th October, resulting in elevated temperature levels of 36 °C. Thus, the proposed solution represents an even-span smart greenhouse. This design ensures good ventilation, with even temperature distribution for orchid and coriander cultivation. Taken together, the highlighted results of three environmental parameters (soil moisture, light intensity, and temperature) include: 22.85 °C temperature, 1686 soil moisture, 57.08% humidity, and 785.24 LDR. All data were collected via an ESP-32-based monitoring system. The small-scale prototype was also designed more cost-effectively compared with the prior solution, such as Bluehouse, which involves high operational costs.

RECOMMENDATION

Real-Life Application:

The even-span greenhouse structure is constructed in the 6th of October City, Egypt, due to severe water scarcity and low soil quality for sustainable cultivation. As illustrated in Figure 14, the desert-like conditions are characterized by high temperature levels (approximately 36°C) and extremely low soil moisture (1%). Furthermore, the city experiences low precipitation (nearly 0.1 mm/day); meanwhile, the average is estimated at 7.5 mm/day.



Figure 14: 6th October City (Proposed Location)

The implementation of the even-span structure would emphasize air circulation and ensure uniform temperature distribution. This is primarily via the utilization of two symmetrical spans, identical slopes with equal lengths. The structure is proposed to be constructed with galvanized steel covered with fiberglass (ACF Greenhouses Buying Guide - Help Choosing a Greenhouse, n.d.) due to its durability and UV resistance. The structure is designed with dimensions of 44.0 × 7.0 m along a height of 2.0 m (Kwon et al., 2015).

Drip Irrigation System

It is recommended to use drip irrigation systems due to their high efficiency in delivering water directly to the plant root zone in a controlled and precise manner. This method significantly improves the efficiency of water usage by reducing evaporation, surface runoff, and unnecessary water distribution, making it highly appropriate for arid environments such as 6th of October City, where water conservation is important, due to a lack of water sources. The system operates through a network of small tubes and emitters that release water drop by drop, which relies on programmed control signals from the loop circuit, which can be integrated with soil moisture, through which the sensor detects the level of the soil moisture, and then sends signals for the loop to allow the work of the system if it is required. This closed feedback mechanism helps maintain optimal soil moisture levels, enhancing plant growth stability and reducing stress caused by overwatering or underwatering. However, it is hard to apply this to the idea of the project because of the highly complex tubing infrastructure, precise clog-free maintenance, and stable, continuous water pressure.

Semiconductor LED Chips

It is recommended to use the Semiconductor LED Chips, due to their importance in providing the appropriate spectrum wavelengths according to the three phases of plant life, which are 1-seedling phase, 2-vegetative phase, 3-flowering phase, and it is a strong edit due to the enhancement leaf and stem development, improving flowering and fruiting efficiency, reducing wasted energy compared to normal white light, and allowing plants to grow in optimized, controlled environments regardless of natural sunlight. The loop circuit is preprogrammed with inputs for the plant's phases and properties, and these inputs provide a signal to the current, and finally to the semiconductor LED material, such as Gallium Nitride, to emit the desired wavelength for each phase. Finally, it can't be applied in the prototype's enclosed system, because of the high cost that ranges between 10000 and 25000 pounds for the small scale of the prototype.

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